

COMPASS Synoptic CB: Sulfide

Sept 2026 Maryland Samples

2026-10-02

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed
sample_month = "Sept"
sample_year = "2026"
Sample_Month_Year = "202609"
Run_Date = as.Date(c("10/01/2026"), format = "%m/%d/%Y")
Run_by = "Zoe Read" #Instrument user
Script_run_by = "Zoe Read" #Code user

Run_notes="Dups and matrix checks look good, spikes look bad. " #any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE, pattern = "Plate.*20261001\\.csv")
(Raw_plates)

## [1] "Raw Data/COMPASS_H2S_Sept2026_Plate1_20261001.csv"
## [2] "Raw Data/COMPASS_H2S_Sept2026_Plate2_20261001.csv"
## [3] "Raw Data/COMPASS_H2S_Sept2026_Plate3_20261001.csv"
## [4] "Raw Data/COMPASS_H2S_Sept2026_Plate4_20261001.csv"

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

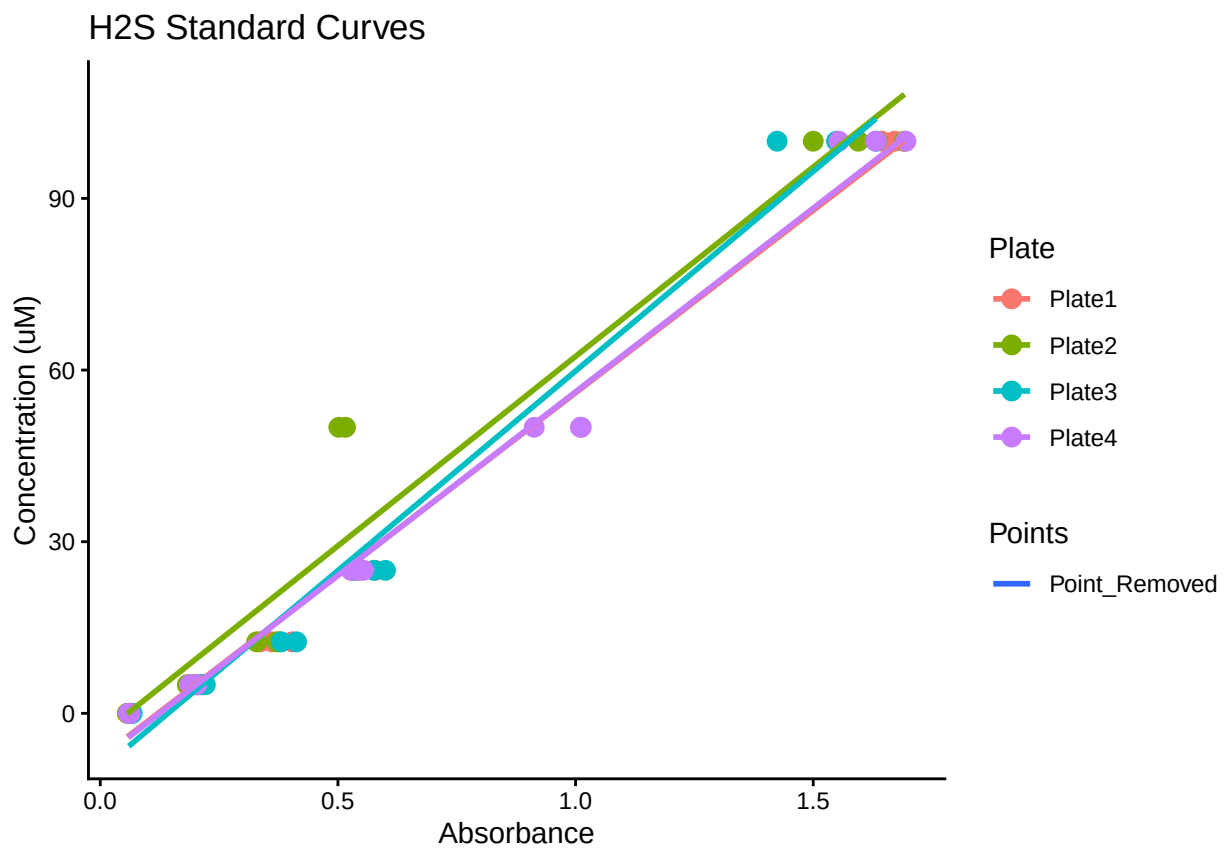
# Metadata path
Raw_Metadata <- "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_SynopticCB_H2S_202609_MD_processed.csv"
rerun_path <- "Processed Data/COMPASS_SynopticCB_H2S_202609_MD_reruns.csv"

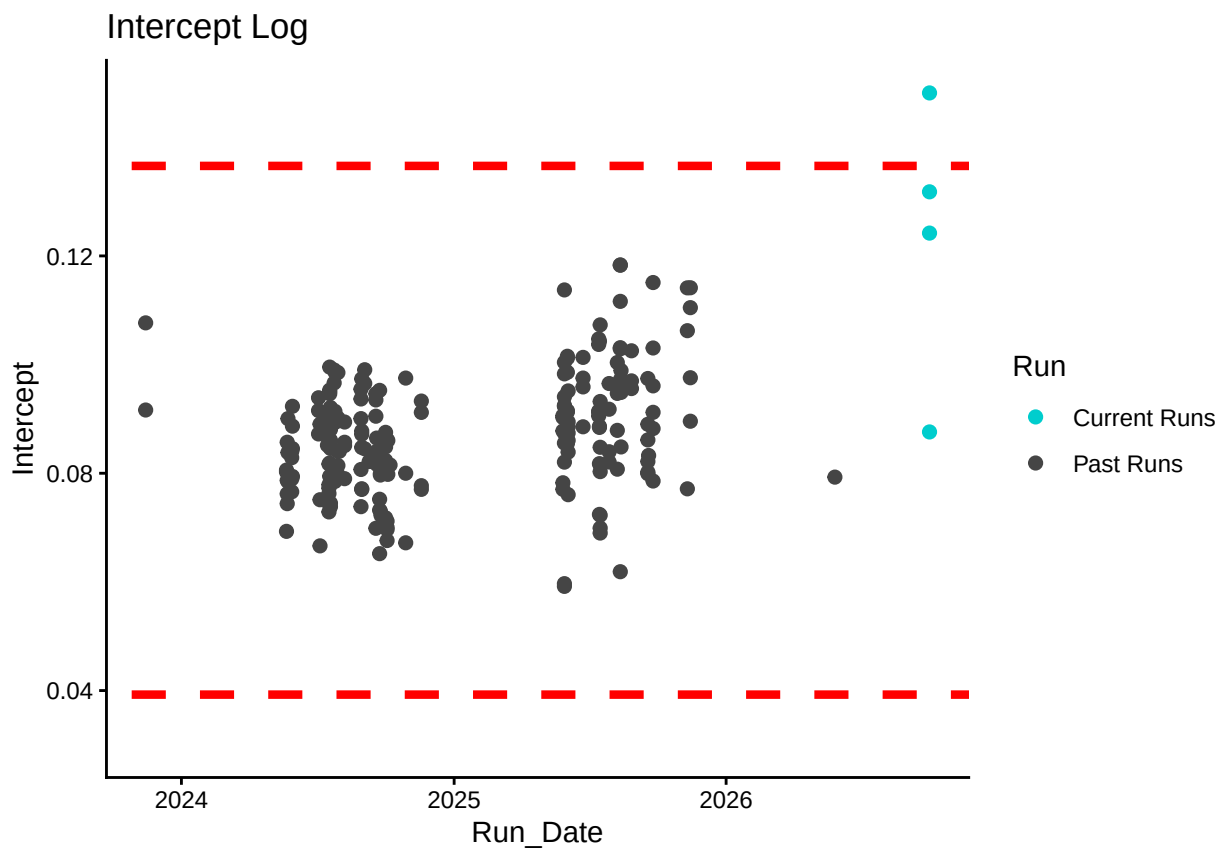
##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary

## Joining with 'by = join_by(Plate, IDs)'
```

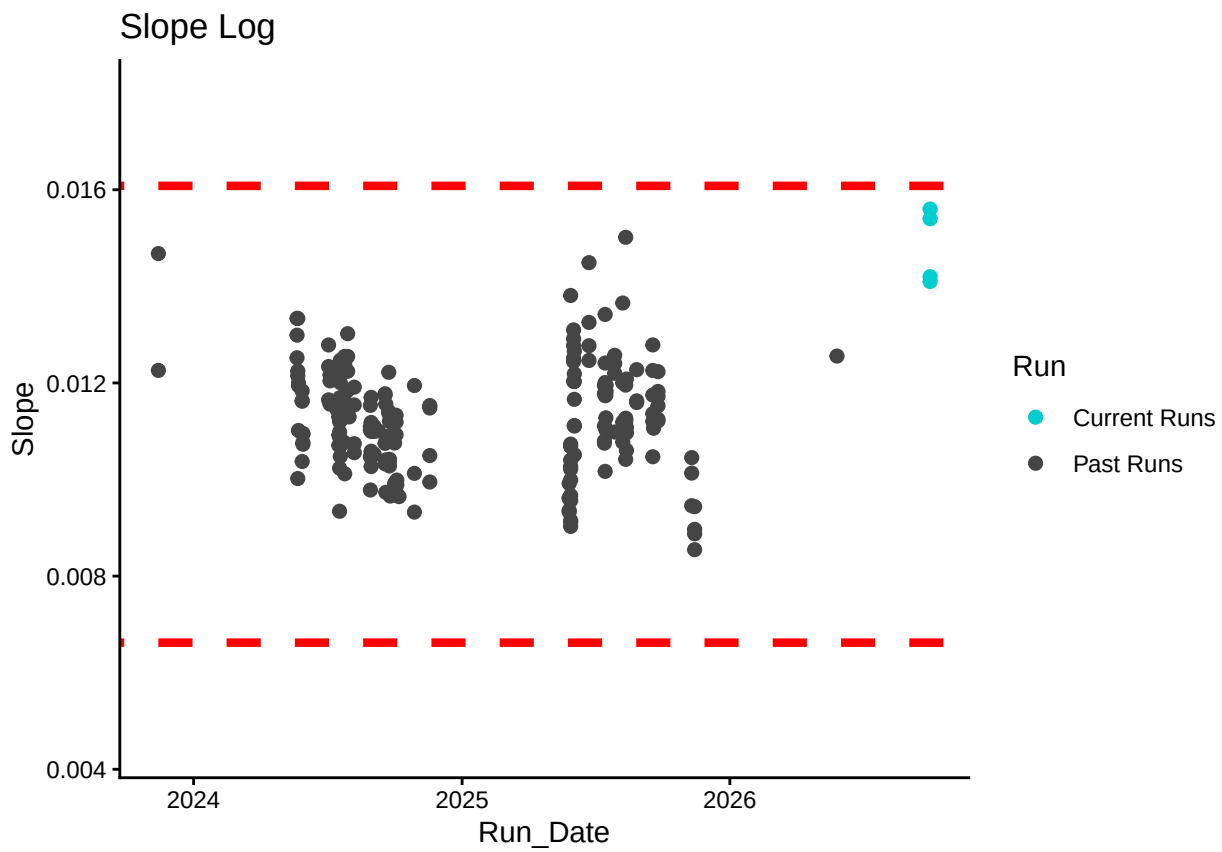
Plot Standard Curves



```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
## Warning: Removed 11 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

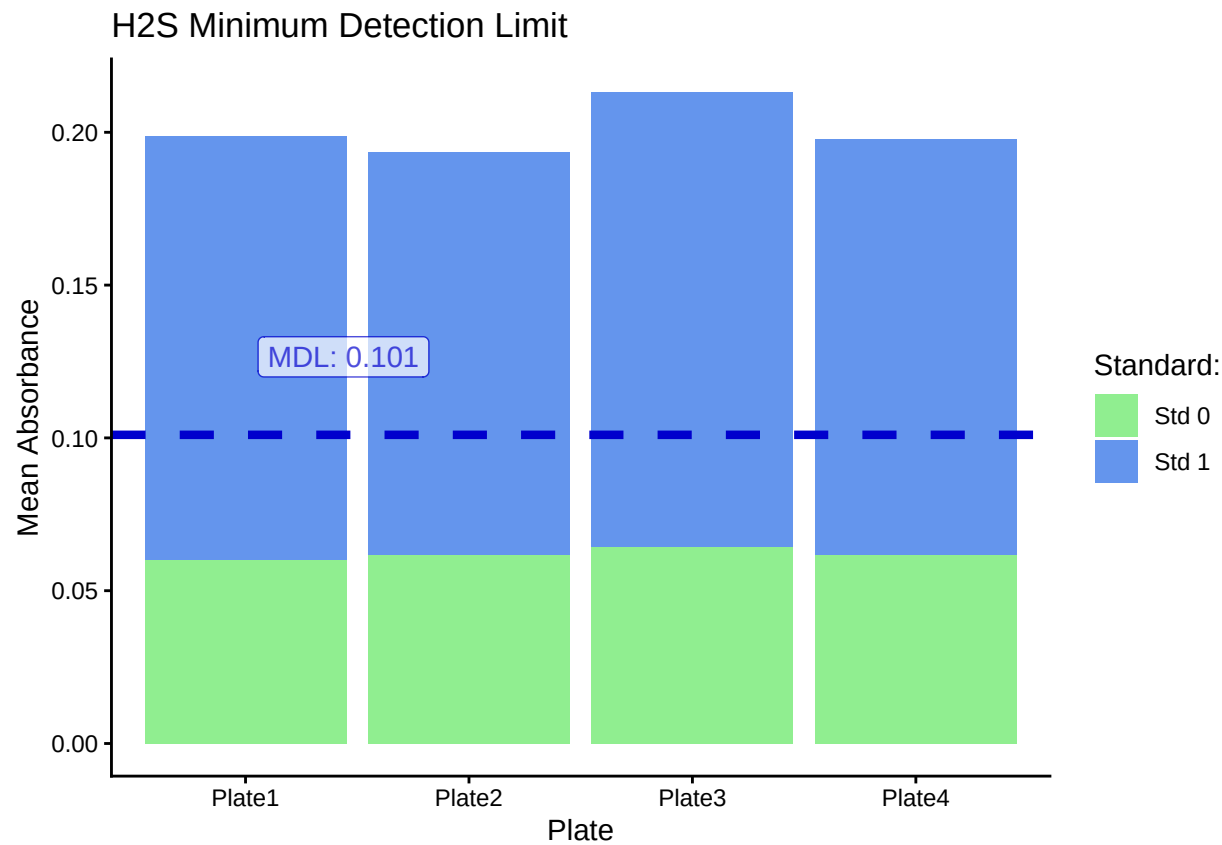


Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Bad	CV_Good
Plate3	R2_Bad	CV_Good
Plate4	R2_Good	CV_Good

Plate	Curve_RMVD
Plate1	KEPT
Plate2	REMOVED
Plate3	REMOVED
Plate4	KEPT

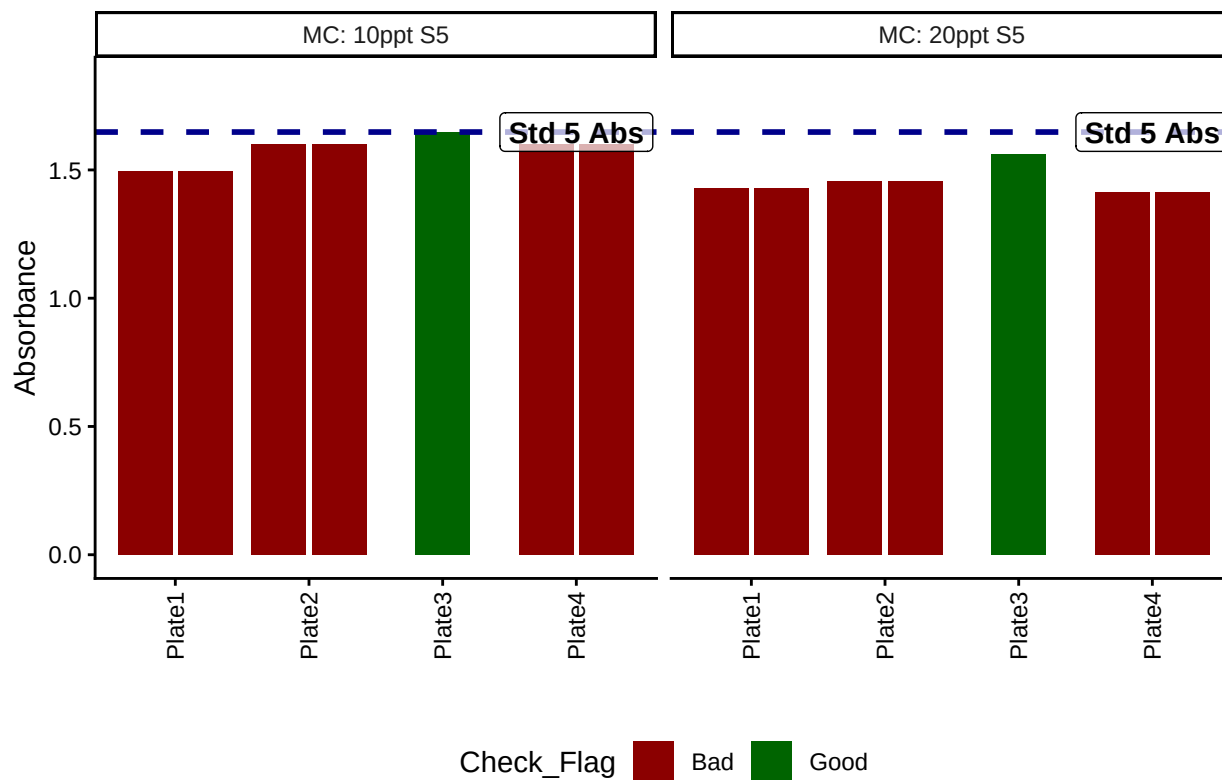
Method Detection Limit



Compare Check Standards to Standards

```
## Warning in left_join(H2S_Chks_bind, Chk_flags_plot, by = join_by(Std_ID, : Detected an unexpected many-
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 1 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many" to silence this warning.
```

Check Stds: H2S with Confidence Interval



Display Any Check Flags

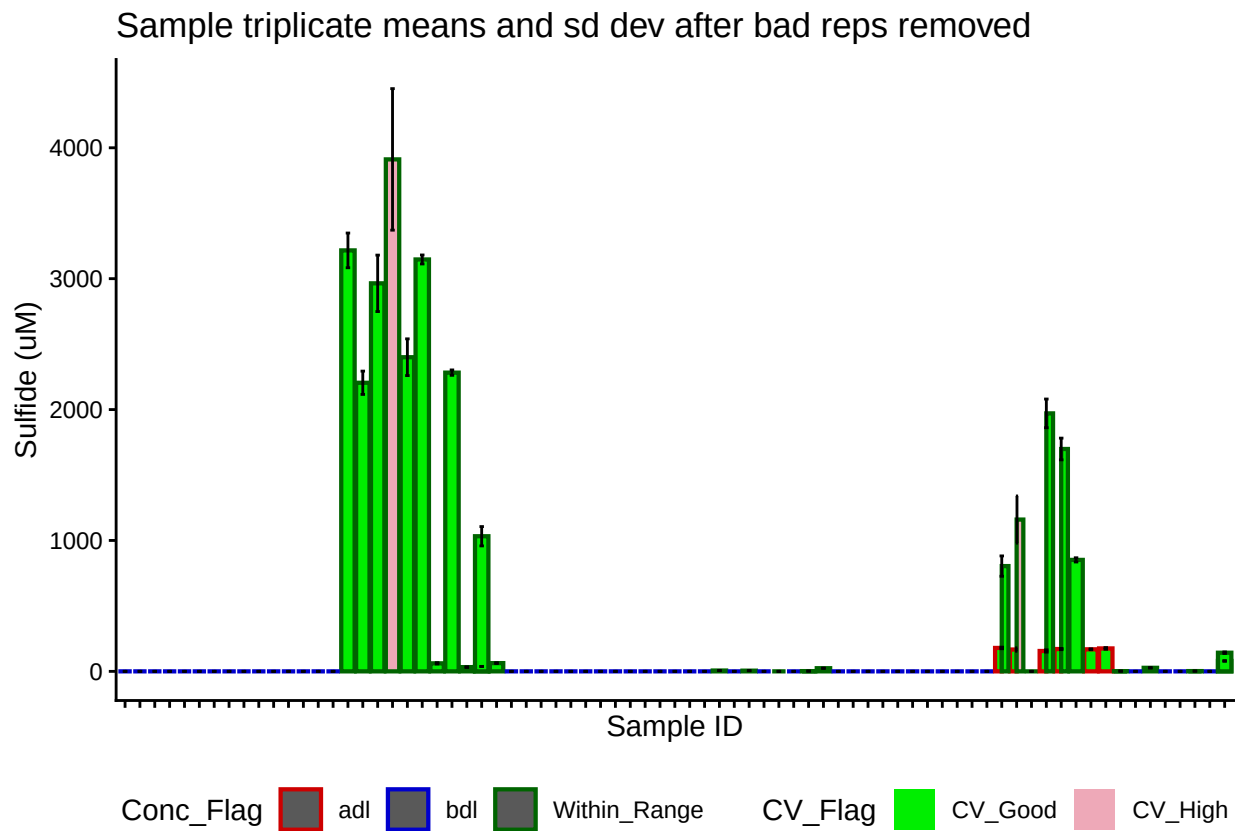
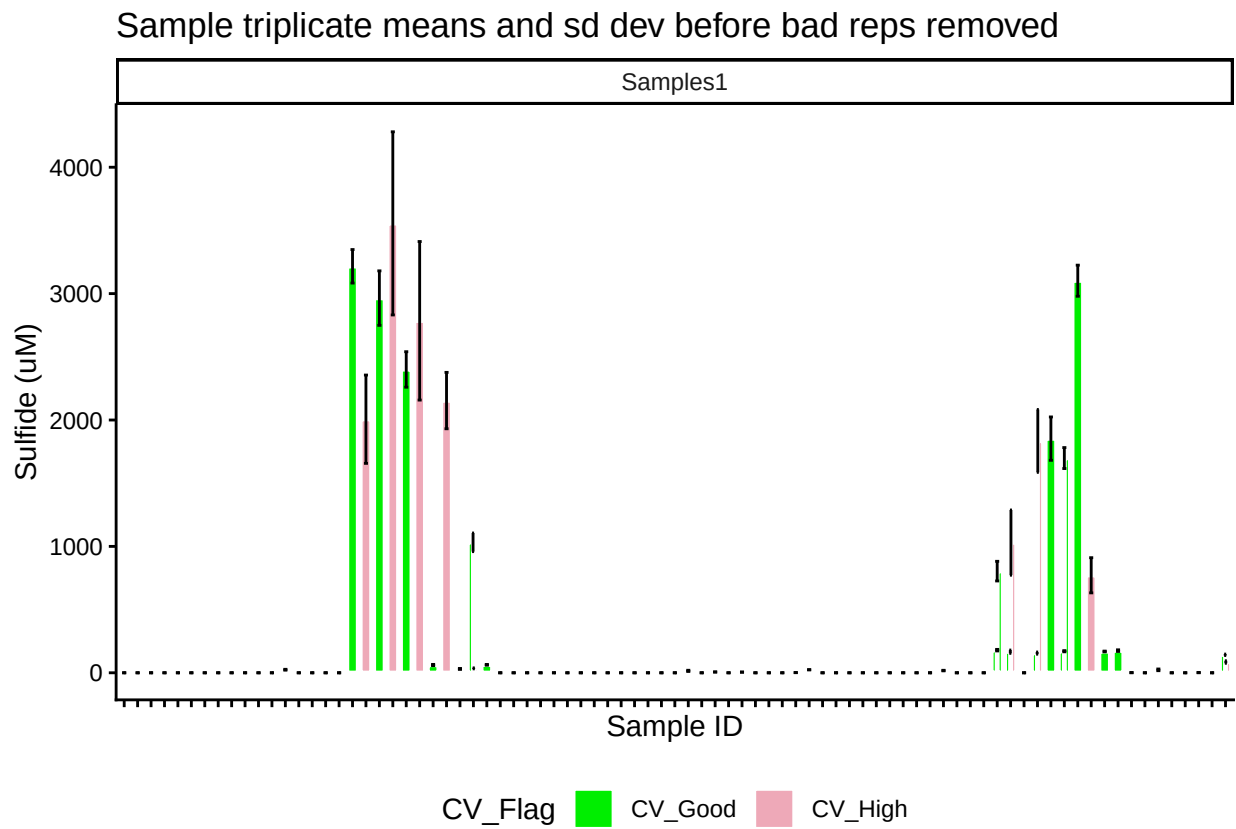
Plate	CHKs_Flag
Plate1	MC: 10ppt S5 Chk_Bad
Plate1	MC: 20ppt S5 Chk_Bad
Plate2	MC: 10ppt S5 Chk_Bad
Plate2	MC: 20ppt S5 Chk_Bad
Plate4	MC: 10ppt S5 Chk_Bad
Plate4	MC: 20ppt S5 Chk_Bad

Plate	CHK_CV_Flags
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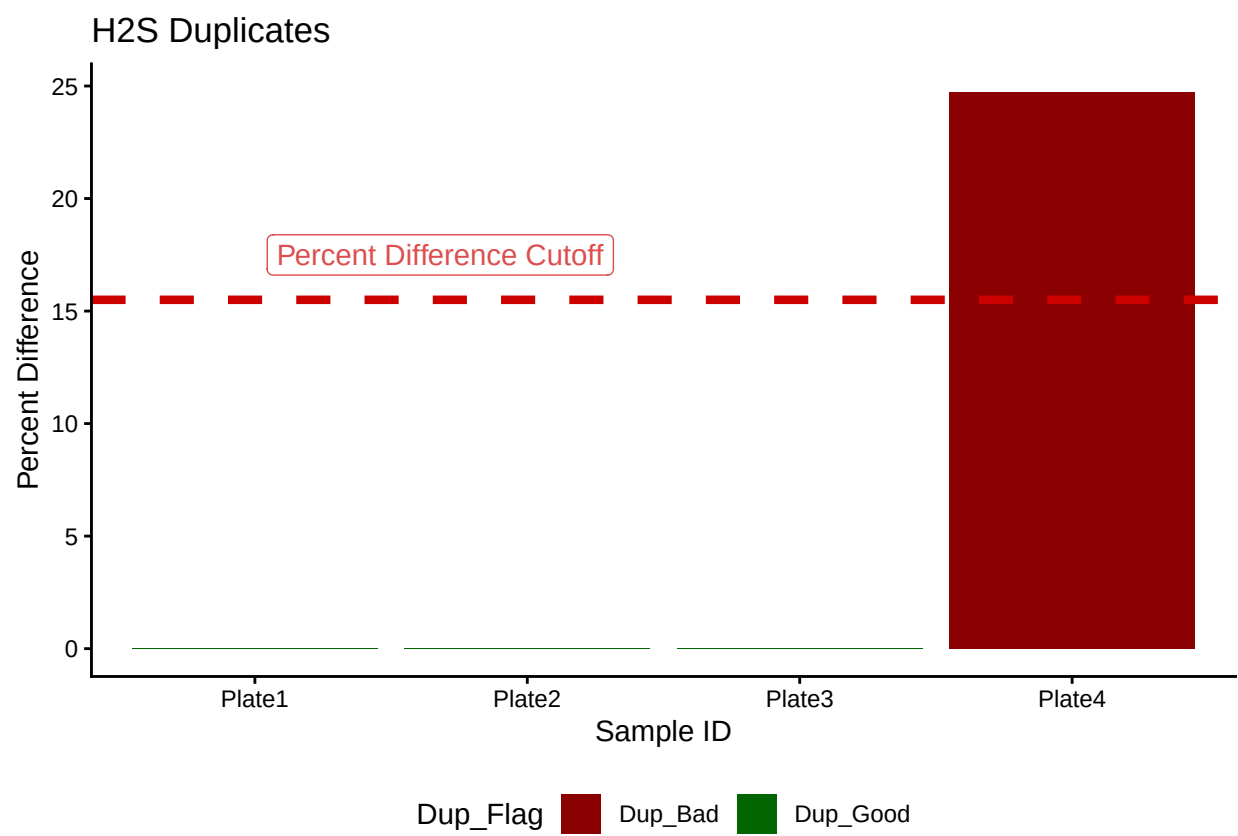
##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

Plot Samples



Duplicate QAQC

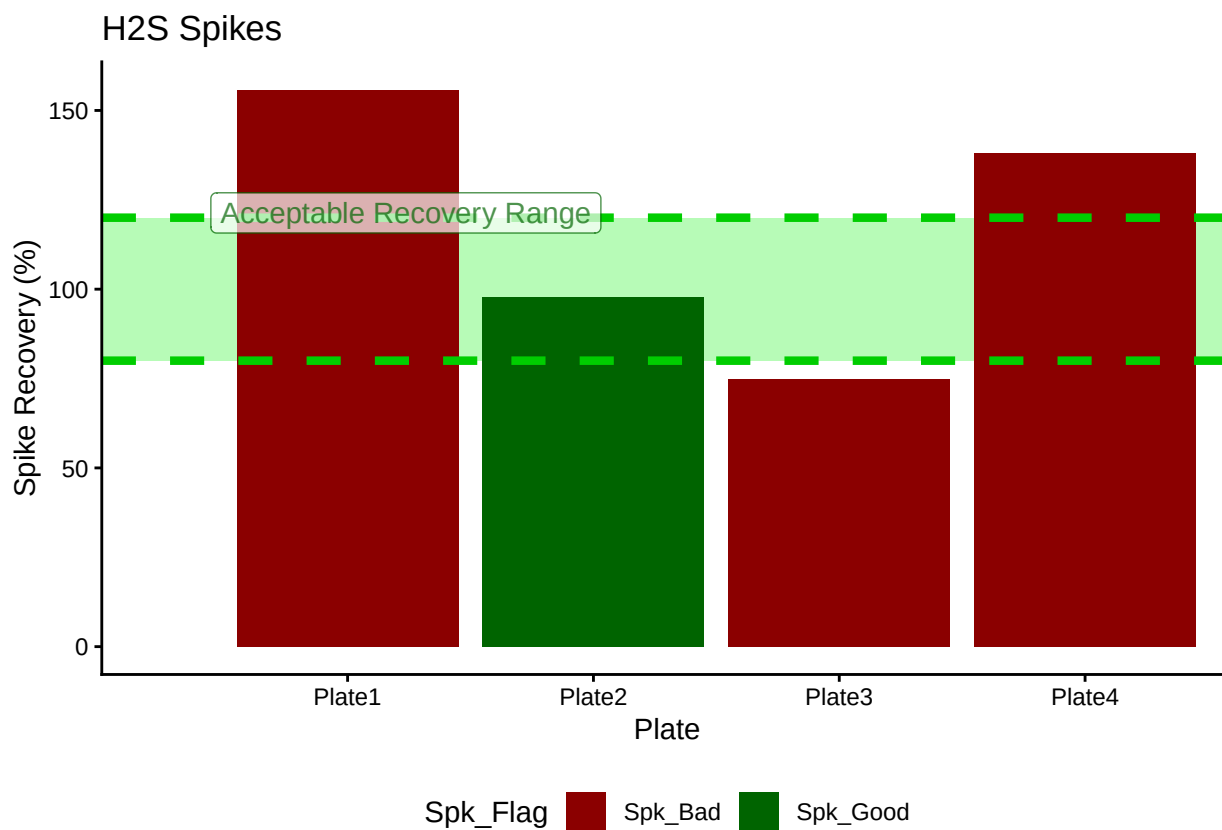


Display Dup Flags

[1] ">60% Dups Pass"

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Good
Plate4	Dup_Bad

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Good
Plate3	Spk_Bad
Plate4	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

```
##Add Flags
```

```
##Remove TEMPEST and FATE samples
```

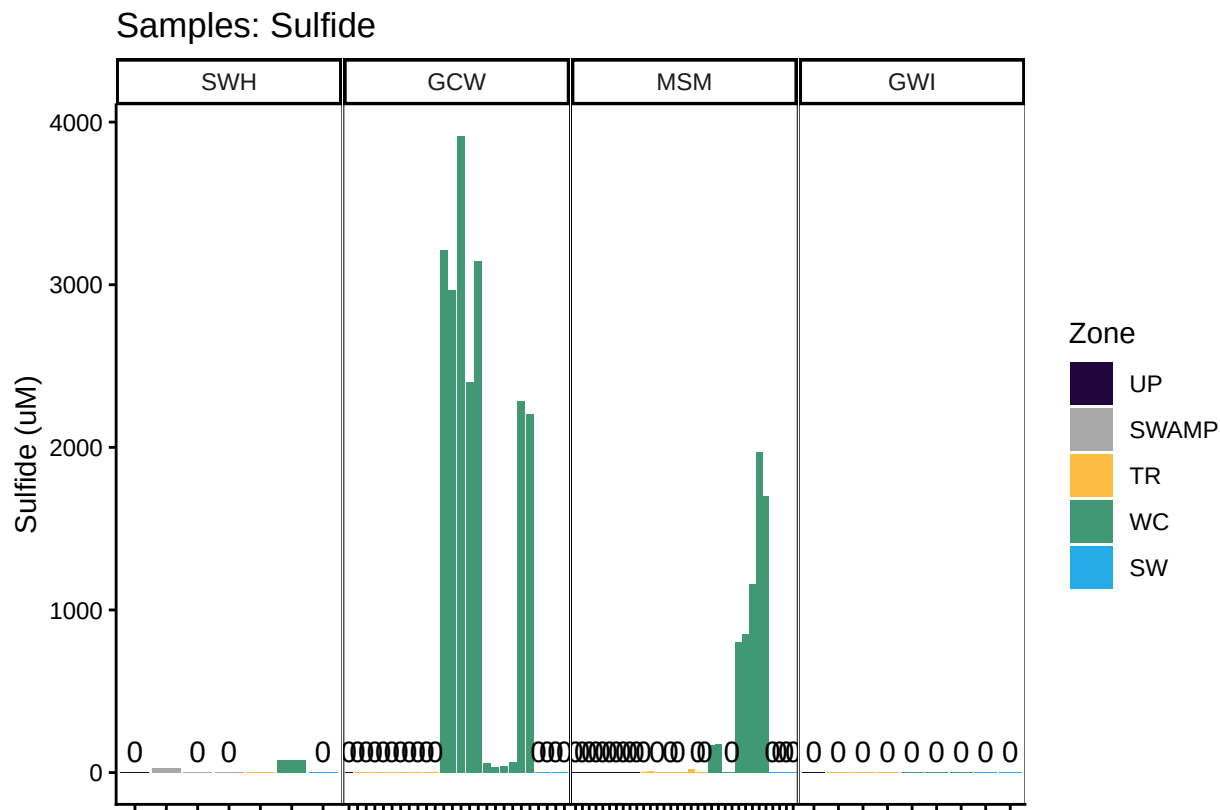
```
Sample_data_IDs <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "TMP")) %>%
  filter(!str_detect(Sample_ID, "FATE")) %>%
  filter(!str_detect(Sample_ID, "dup")) %>%
  filter(!str_detect(Sample_ID, "spk"))
```


Check to see if samples run match metadata & merge info

Some sample IDs are missing from metadata.

```
## [1] "SWH_202609_SWAMP_LYSA_20CM" "GCW_202609_TR_LYSA_10CM"
## [3] "GCW_202609_TR_LYSC_10CM"    "SWH_202609_SWAMP_LYSC_20CM"
## [5] "GCW_202609_TR_LYSA_20CM"    "GCW_202609_TR_LYSC_20CM"
## [7] "GWI_202609_TR_LYSA_10CM"    "GCW_202609_TR_LYSA_45CM"
## [9] "GCW_202609_TR_LYSC_45CM"    "GWI_202609_TR_LYSA_20CM"
## [11] "GCW_202609_TR_LYSB_10CM"    "GCW_202609_WC_LYSA_10CM"
## [13] "GWI_202609_SW_A"            "GCW_202609_TR_LYSB_20CM"
## [15] "GCW_202609_WC_LYSA_20CM"    "GWI_202609_WC_LYSA_20CM"
## [17] "GCW_202609_WC_LYSA_45CM"    "GWI_202609_WC_LYSA_45CM"
## [19] "GCW_202609_TR_LYSB_45CM"    "GCW_202609_WC_LYSB_10CM"
## [21] "GCW_202609_UP_LYSA_10CM"    "GCW_202609_WC_LYSB_20CM"
## [23] "GCW_202609_WC_LYSB_45CM"    "MSM_202609_UP_LYSA_20CM"
## [25] "MSM_202609_UP_LYSC_20CM"    "GCW_202609_WC_LYSC_10CM"
## [27] "MSM_202609_UP_LYSA_45CM"    "MSM_202609_UP_LYSC_45CM"
## [29] "GCW_202609_WC_LYSC_20CM"    "MSM_202609_UP_LYSB_10CM"
## [31] "MSM_202609_TR_LYSA_10CM"    "GCW_202609_WC_LYSC_45CM"
## [33] "MSM_202609_UP_LYSB_20CM"    "MSM_202609_TR_LYSA_20CM"
## [35] "GCW_202609_SW_A"            "MSM_202609_UP_LYSB_45CM"
## [37] "MSM_202609_TR_LYSA_45CM"    "GCW_202609_SW_B"
## [39] "MSM_202609_TR_LYSB_10CM"    "GCW_202609_SW_C"
## [41] "MSM_202609_UP_LYSC_10CM"    "MSM_202609_TR_LYSB_20CM"
## [43] "MSM_202609_UP_LYSA_10CM"    "MSM_202609_TR_LYSB_45CM"
## [45] "MSM_202609_TR_LYSC_10CM"    "MSM_202609_WC_LYSC_10CM"
## [47] "MSM_202609_TR_LYSC_20CM"    "MSM_202609_WC_LYSC_20CM"
## [49] "MSM_202609_TR_LYSC_45CM"    "MSM_202609_WC_LYSC_45CM"
## [51] "MSM_202609_SW_A"            "SWH_202609_UP_LYSA_20CM"
## [53] "MSM_202609_SW_B"            "SWH_202609_TR_LYSA_10CM"
## [55] "MSM_202609_WC_LYSA_45CM"    "SWH_202609_SWAMP_LYSC_10CM"
## [57] "MSM_202609_SW_C"            "SWH_202609_WC_LYSC_10CM"
## [59] "SWH_202609_SW_B"            "MSM_202609_WC_LYSA_10CM"
## [61] "MSM_202609_WC_LYSB_20CM"    "MSM_202609_WC_LYSA_20CM"
## [63] "GWI_202609_UP_LYSA_20CM"    "MSM_202609_WC_LYSB_10CM"
## [65] "GWI_202609_WC_LYSC_20CM"
```

Visualize Data by Plot



###END